

Лабораторная работа №5

Конфигурация VLAN

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1 Информация

Раздел 1

Информация

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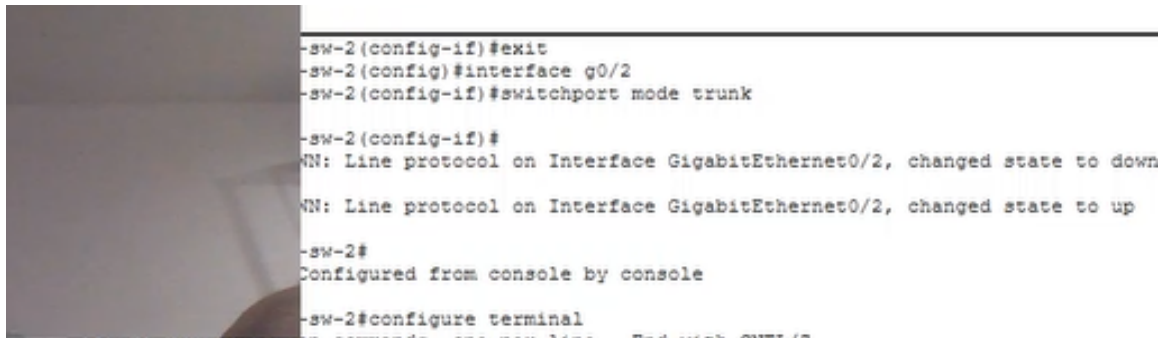


Цель работы

Получить основные навыки по настройке VLAN на коммутаторах сети.

Выполнение работы

Используя приведённую в лабораторной работе последовательность команд из примера по конфигурации Trunk-порта на интерфейсе g0/1 коммутатора msk-donskaya-sw-1, настроим Trunk-порты на соответствующих интерфейсах всех коммутаторов (рис. #fig:002 – #fig:006):

A screenshot of a terminal window with a dark background and light-colored text. The text shows the configuration of a Trunk port on a switch named 'sw-2'. The commands entered are: 'exit' to leave the interface configuration, 'interface g0/2' to enter the configuration for GigabitEthernet0/2, and 'switchport mode trunk' to set the port to Trunk mode. The prompt changes from '(config-if)' to '(config)' and then back to the root prompt '#'. Status messages indicate the line protocol on the interface has changed from down to up. Finally, the user enters 'configure terminal' to return to global configuration mode.

```
-sw-2(config-if)#exit
-sw-2(config)#interface g0/2
-sw-2(config-if)#switchport mode trunk

-sw-2(config-if)#
NN: Line protocol on Interface GigabitEthernet0/2, changed state to down
NN: Line protocol on Interface GigabitEthernet0/2, changed state to up

-sw-2#
Configured from console by console

-sw-2#configure terminal
```

Рисунок 1: Настройка Trunk-портов на коммутаторе msk-donskaya-agko-sw-2

msk-donskaya-agko-sw-3>

msk-donskaya-agko-sw-3>enable

Password:

msk-donskaya-agko-sw-3#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

msk-donskaya-agko-sw-3(config)#interface g0/1

msk-donskaya-agko-sw-3(config-if)#switchport mode trunk

msk-donskaya-agko-sw-3(config-if)#^Z

msk-donskaya-agko-sw-3#

4SYS-5-CONFIG_I: Configured from console by console

write memory

Building configuration...

[OK]

msk-donskaya-agko-sw-3#

Рисунок 2: Настройка Trunk-портов на коммутаторе msk-donskaya-agko-sw-3

Password:

```
msk-donskaya-agko-sw-4>enable
```

Password:

```
msk-donskaya-agko-sw-4#conf terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
msk-donskaya-agko-sw-4(config)#interface g0/2
```

```
msk-donskaya-agko-sw-4(config-if)#switchport mode trunk
```

```
msk-donskaya-agko-sw-4(config-if)#^Z
```

```
msk-donskaya-agko-sw-4#
```

```
4SYS-5-CONFIG_I: Configured from console by console
```

```
WR M
```

```
Building configuration...
```

```
[OK]
```

```
msk-donskaya-agko-sw-4#
```

Рисунок 3: Настройка Trunk-портов на коммутаторе msk-donskaya-agko-sw-4

Password:

msk-pavlovskaya-agko-1>enable

Password:

msk-pavlovskaya-agko-1#interface f0/24

% Invalid input detected at '^' marker.

msk-pavlovskaya-agko-1#interface f0/24

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-donskaya-agko-sw-1 FastEthe

msk-pavlovskaya-agko-1#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

msk-pavlovskaya-agko-1(config)#terminal f0/24

% Invalid input detected at '^' marker.

msk-pavlovskaya-agko-1(config)#interface f0/24

msk-pavlovskaya-agko-1(config-if)#switchport mode trunk

msk-pavlovskaya-agko-1(config-if)#

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

^Z

msk-pavlovskaya-agko-1#

%SYS-5-CONFIG_I: Configured from console by console

wr m

Copy

Paste

Далее настроим коммутатор msk-donskaya-agko-sw-1 как VTP-сервер и пропишем на нём номера и названия VLAN (рис. #fig:007):

```
msk-donskaya-agko-sw-1>enable
Password:
msk-donskaya-agko-sw-1#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-agko-sw-1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with msk-
pavlovskaya-agko-1 FastEthernet0/24 (1).
vtp mode server
Setting device to VTP SERVER mode.
msk-donskaya-agko-sw-1(config)#vtp domain donsкаya
Domain name already set to donsкаya.
msk-donskaya-agko-sw-1(config)#vtp password cisco
Password already set to cisco
msk-donskaya-agko-sw-1(config)#vlan 2
msk-donskaya-agko-sw-1(config-vlan)#name management
msk-donskaya-agko-sw-1(config-vlan)#vlan 3
msk-donskaya-agko-sw-1(config-vlan)#
msk-donskaya-agko-sw-1(config-vlan)#name servers
msk-donskaya-agko-sw-1(config-vlan)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (101), with msk-
pavlovskaya-agko-1 FastEthernet0/24 (1).
vlan 101
msk-donskaya-agko-sw-1(config-vlan)#vlan 101
msk-donskaya-agko-sw-1(config-vlan)#name dk
msk-donskaya-agko-sw-1(config-vlan)#vlan 102
msk-donskaya-agko-sw-1(config-vlan)#name departaments
msk-donskaya-agko-sw-1(config-vlan)#vlan 103
msk-donskaya-agko-sw-1(config-vlan)#name adm
msk-donskaya-agko-sw-1(config-vlan)#vlan 104
msk-donskaya-agko-sw-1(config-vlan)#
```

Теперь настроим коммутаторы msk-donskaya-agko-sw-2, msk-donskaya-agko-sw-3, msk-donskaya-agko-sw-4 и msk-pavlovskaya-agko-sw-1 как VTP-клиенты и на интерфейсах укажем принадлежность к VLAN (рис. #fig:008 – #fig:011):

```
% Invalid input detected at '^' marker.
```

```
msk-donskaya-agko-sw-2(config)#vtp mode server
```

```
Setting device to VTP SERVER mode.
```

```
msk-donskaya-agko-sw-2(config)#interface range f0/1-2
```

```
msk-donskaya-agko-sw-2(config-if-range)#switchport mode access
```

```
msk-donskaya-agko-sw-2(config-if-range)#switchport access vlan 101
```

```
msk-donskaya-agko-sw-2(config-if-range)#switchport access vlan 3
```

```
msk-donskaya-agko-sw-2(config-if-range)^Z
```

```
msk-donskaya-agko-sw-2#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
wr m
```

```
Building configuration...
```

```
[OK]
```

```
msk-donskaya-agko-sw-2#
```

```
msk-donskaya-agko-sw-3#  
msk-donskaya-agko-sw-3#conf terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
msk-donskaya-agko-sw-3(config)#vtp mode client  
Setting device to VTP CLIENT mode.  
msk-donskaya-agko-sw-3(config)#interface f0/1  
msk-donskaya-agko-sw-3(config-if)#switchport mode access  
msk-donskaya-agko-sw-3(config-if)#switchport access vlan 3  
msk-donskaya-agko-sw-3(config-if)#^Z  
msk-donskaya-agko-sw-3#  
%SYS-5-CONFIG_I: Configured from console by console
```

Рисунок 7: Настройка коммутатора msk-donskaya-agko-sw-3 как VTP-клиента и указание принадлежности к VLAN


```
msk-donskaya-agko-sw-4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-agko-sw-4(config)#interface g0/2
msk-donskaya-agko-sw-4(config-if)#switchport mode trunk
msk-donskaya-agko-sw-4(config-if)#^Z
msk-donskaya-agko-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-donskaya-agko-sw-4#
msk-donskaya-agko-sw-4#conf terminal
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-agko-sw-4(config)#vtp mode client
Setting device to VTP CLIENT mode.
msk-donskaya-agko-sw-4(config)#interface range f0/1 - 5
msk-donskaya-agko-sw-4(config-if-range)#switchport mode access
msk-donskaya-agko-sw-4(config-if-range)#switchport access vlan 101
msk-donskaya-agko-sw-4(config-if-range)#switchport mode access
msk-donskaya-agko-sw-4(config-if-range)#interface range f0/6 - 10
msk-donskaya-agko-sw-4(config-if-range)#switchport mode access
msk-donskaya-agko-sw-4(config-if-range)#switchport access vlan 102
msk-donskaya-agko-sw-4(config-if-range)#interface range f0/11 - 15
msk-donskaya-agko-sw-4(config-if-range)#switchport mode access
msk-donskaya-agko-sw-4(config-if-range)#switchport access vlan 103
msk-donskaya-agko-sw-4(config-if-range)#interface range f0/16 - 24
msk-donskaya-agko-sw-4(config-if-range)#switchport mode access
msk-donskaya-agko-sw-4(config-if-range)#switchport access vlan 104
msk-donskaya-agko-sw-4(config-if-range)#^Z
msk-donskaya-agko-sw-4#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-donskaya-agko-sw-4#
```

```
msk-pavlovskaya-agko-1(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-
donskaya-agko-sw-1 FastEthernet0/1 (101).
vtp mode client
Device mode already VTP CLIENT.
msk-pavlovskaya-agko-1(config)#interface range f0/1 - 15
msk-pavlovskaya-agko-1(config-if-range)#switchport mode access
msk-pavlovskaya-agko-1(config-if-range)#switchport access vlan 101
msk-pavlovskaya-agko-1(config-if-range)#interface f0/20
msk-pavlovskaya-agko-1(config-if)#switchport mode access
msk-pavlovskaya-agko-1(config-if)#switchport access vlan 104
msk-pavlovskaya-agko-1(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/24 (1), with msk-
donskaya-agko-sw-1 FastEthernet0/1 (101).

msk-pavlovskaya-agko-1(config-if)#
msk-pavlovskaya-agko-1#
%SYS-5-CONFIG_I: Configured from console by console
wr m
Building configuration...
[OK]
msk-pavlovskaya-agko-1#
```

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Рисунок 9: Настройка коммутатора msk-pavlovskaya-agko-sw-1 как VTP-клиента и указание принадлежности к VLAN

Затем требуется указать статические IP-адреса на оконечных устройствах (рис. #fig:012 – #fig:013):

The screenshot shows a network configuration window for a device named 'adm-donskaya-1'. The 'Interfaces' section is set to 'FastEthernet0'. Under the 'Gateway/DNS IPv4' section, the 'Static' option is selected. The 'Default Gateway' is configured as '10.128.0.26' and the 'DNS Server' is configured as '255.255.255.0'. Navigation icons are visible at the bottom right of the window.

Display Name	adm-donskaya-1
Interfaces	FastEthernet0
Gateway/DNS IPv4	
☒	DHCP
☒	Static
Default Gateway	10.128.0.26
DNS Server	255.255.255.0

FastEthernet0

Port Status ☒ On
 Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto
 Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
 MAC Address 0002.1625.8BE4

IP Configuration

☒ DHCP
☒ Static

IPv4 Address 10.128.0.3

Subnet Mask 255.255.0.0

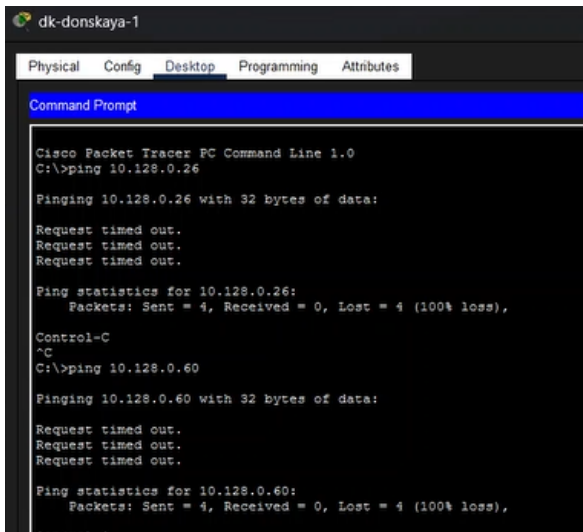
IPv6 Configuration

☒ Automatic
☒ Static

IPv6 Address

Link Local Address: FE80::202:16FF:FE25:8BE4

После указания статических IP-адресов на оконечных устройствах проверим с помощью команды `ping` доступность устройств, принадлежащих одному VLAN, и недоступность устройств, принадлежащих разным VLAN (рис. #fig:014):



```
dk-donskaya-1
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.26

Pinging 10.128.0.26 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.0.26:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Control-C
^C
C:\>ping 10.128.0.60

Pinging 10.128.0.60 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.128.0.60:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

В ходе выполнения лабораторной работы мы получили основные навыки по настройке VLAN на коммутаторах сети.

